

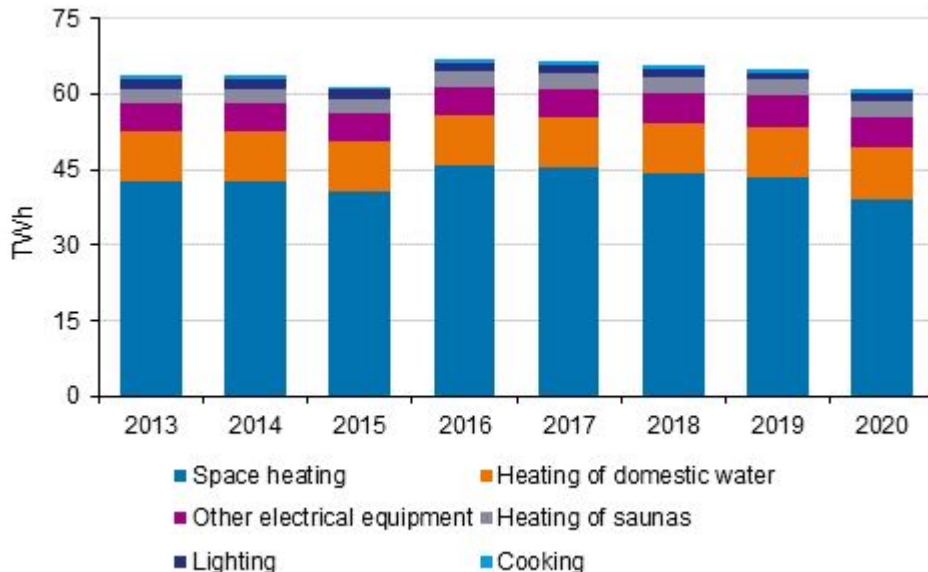
Power Surge

Electricity



Have you ever felt that maybe you were spending too much on electricity? Or maybe you don't feel exactly safe near your electrical appliances?

While people monitor how much electricity they consume each day, we often overlook the hidden dangers behind those numbers. High power consumption isn't just a blow to your wallet, it can even be a fatal blow to your devices!



Our Project



Awareness & Stability

Helps you with understanding expenses involving electricity consumption and gives you a history of consumption

Safety & Prevention

Monitor consumption and defends your devices from power surges.

Our device warns users before levels become critical, and in some extreme cases ceases electricity connections.

Eco-friendly & Energy Efficient

Encourages better energy spending/consumption .

Helps reduce energy waste by sharing in dept data from the outlets..

Features



Device detection

Detects what type of device is plugged in.

Power history and price comparison

Graphs of power and comparison to current electricity prices

Dashboard UI

See power usage and device type.

Safety measures

In case of power surges or strange consumption, shuts off device or sends a warning.

Components

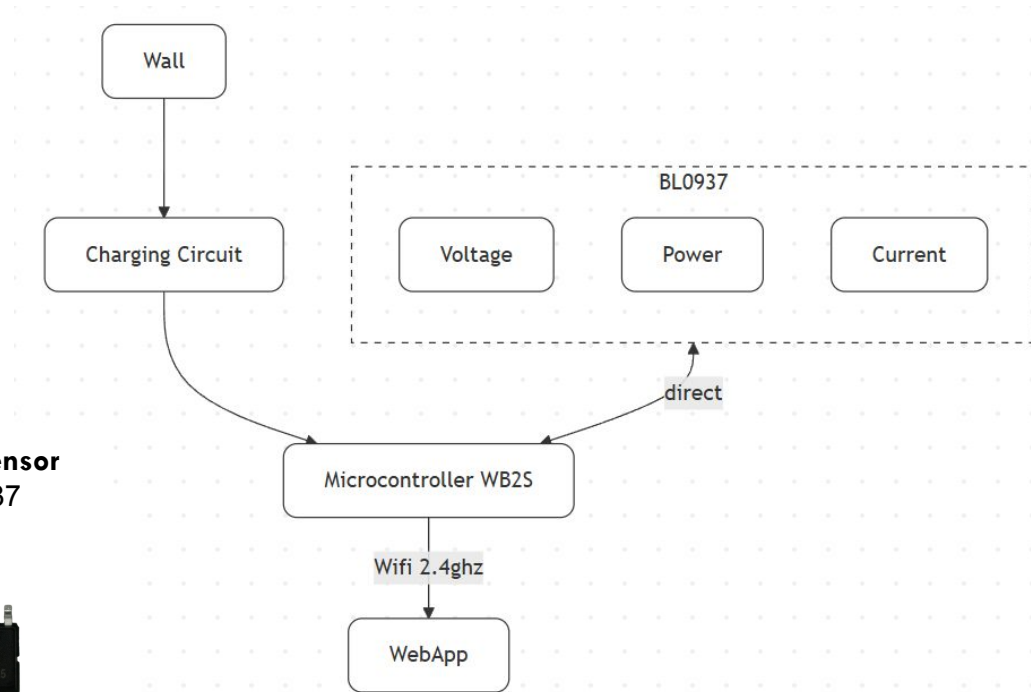
**Microcontroller
WB2S**



Emax Outlet



**Power Sensor
BL0937**

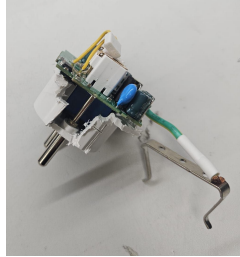


Process (The smart plug)

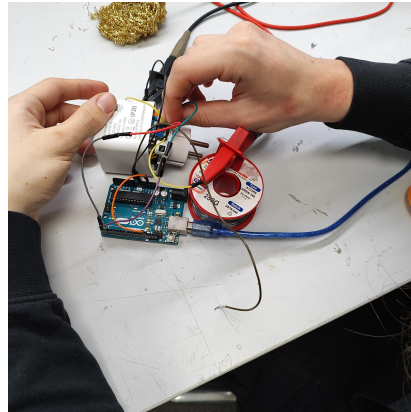
In our first attempt we looked inside the smart plug to understand its components and circuit.

We first flashed the microcontroller to see if it was feasible to change it's firmware,after we ascertained the viability of the project,we took a second outlet and carefully flashed its micro controller(with ESP home),while making sure not to damage anything important.

First attempt:



Flashing Process

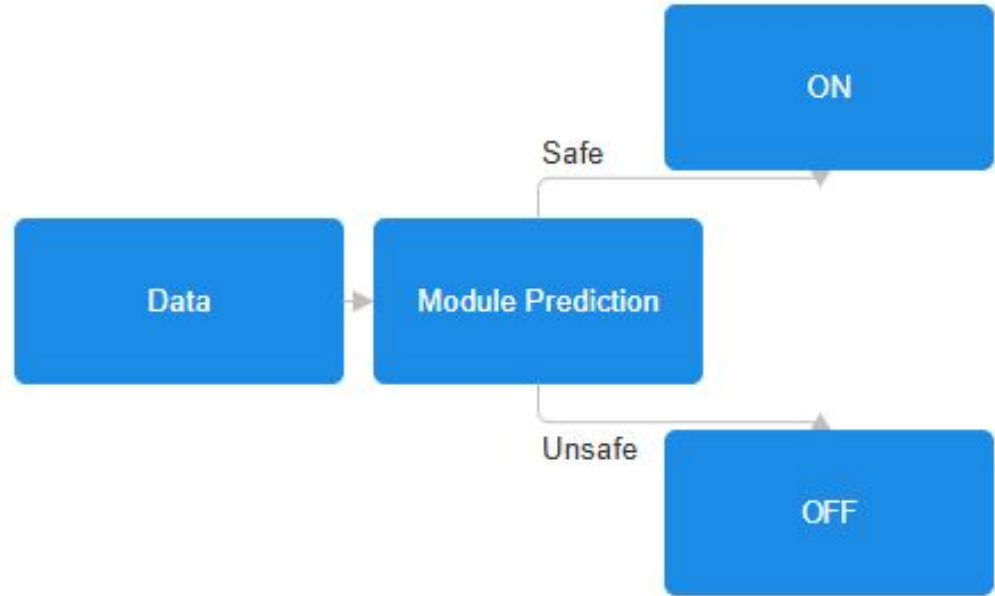


Finished Prototype

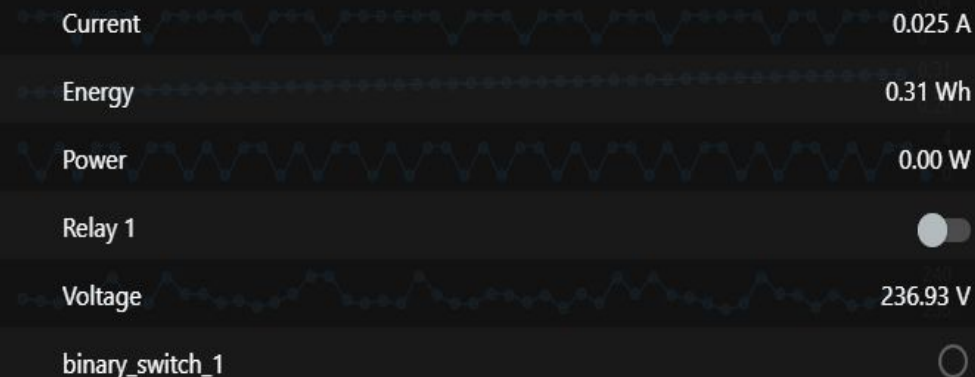


Process (Prediction and webApp):

After installing EspHome we started gathering data for the tensorflow model, the model is what gives us the ability to recognise devices, set their max power or notice any incongruencies with consumption.



Sensor and Control



Diagnostic



LibreTiny Version

v1.12.1+sha.73564e9 on cb2s, compiled at
Mar 29 2026 21:14:11, GCC 10.3.1 (-O1)

WiFi Signal Sensor

-33 dBm

OTA Update

Scegli il file NESSUN FILE SCELTO

UPDATE

Debug Log

Time	Level	Tag	Message
15:49:41	[D]	[sensor:118]	'Power' >> 0.00 W
15:49:41	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:42	[D]	[hlw8012:082]	Got power=3.0W, voltage=204.9V
15:49:42	[D]	[sensor:118]	'Voltage' >> 235.65 V
15:49:42	[D]	[sensor:118]	'Power' >> 3.01 W
15:49:42	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:42	[D]	[hlw8012:082]	Got power=3.0W, voltage=206.3V
15:49:42	[D]	[sensor:118]	'Voltage' >> 237.24 V
15:49:42	[D]	[sensor:118]	'Power' >> 3.01 W
15:49:42	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:42	[D]	[hlw8012:082]	Got power=0.0W, voltage=205.1V
15:49:42	[D]	[sensor:118]	'Voltage' >> 235.81 V
15:49:42	[D]	[sensor:118]	'Power' >> 0.00 W
15:49:42	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:43	[D]	[hlw8012:082]	Got power=3.0W, voltage=205.3V
15:49:43	[D]	[sensor:118]	'Voltage' >> 236.13 V
15:49:43	[D]	[sensor:118]	'Power' >> 3.01 W
15:49:43	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:43	[D]	[hlw8012:082]	Got power=0.0W, voltage=206.3V
15:49:43	[D]	[sensor:118]	'Voltage' >> 237.24 V
15:49:43	[D]	[sensor:118]	'Power' >> 0.00 W
15:49:43	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:44	[D]	[sensor:118]	'Power' >> 3.01 W
15:49:44	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:44	[D]	[hlw8012:076]	Got power=3.0W, current=0.0A
15:49:44	[D]	[sensor:118]	'Current' >> 0.000 A
15:49:44	[D]	[sensor:118]	'Power' >> 3.01 W
15:49:44	[D]	[sensor:118]	'Energy' >> 0.31 Wh
15:49:44	[D]	[hlw8012:076]	Got power=0.0W, current=0.0A